# Chapter 3
# VECTORS

In this chapter we are going to discuss the following topics:

- ❑ **Coordinates Systems**
- ❑ **Vectors & Scalars**
- ❑ **Adding and Subtracting Vectors**
- ❑ **Components of Vectors**
- ❑ **Product of Vectors (Multiplying a Vector by a Scalar, Scalar product, & Vector Product)**

## Recommended Problems:

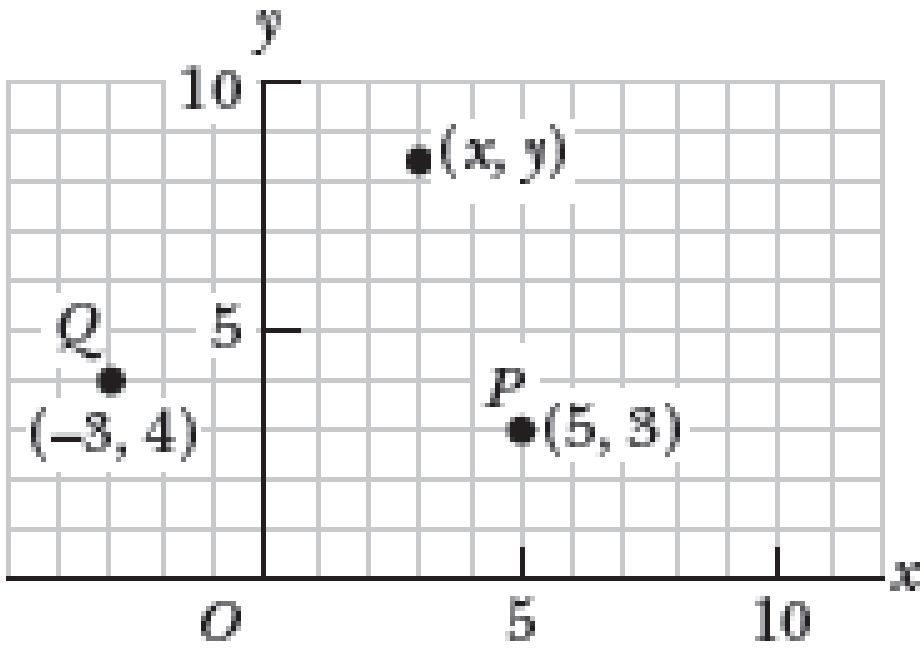
15, 19, 21, 23, 26, 31, 36, 37, 41, 45, 57, 59 (problems from Ch.7)
7, 9, 12, 13 (problems from Ch.11) 1, 3, 7, 10

# 3.1 Coordinates Systems

In physics it is important to define the position of a point in space. In doing so we need a coordinate system.

## Cartesian Coordinates:

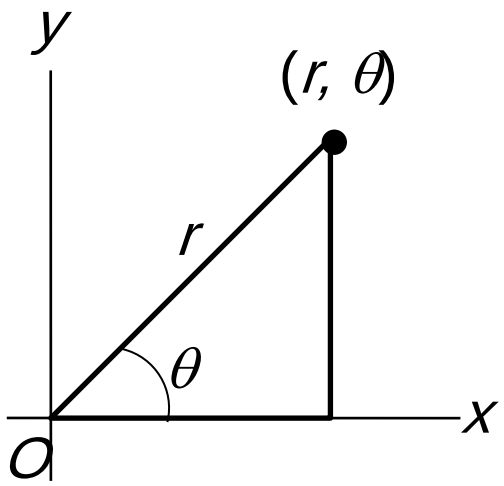
It consists of two perpendicular axes (x-axis, and y-axis) that intersect at a point called the origin. Any point in a plane is assigned two numerical values: x-value which gives the position of the point along the x-axis, and y-value which gives the position of the point along the y-axis.




For 3-dimensional case there is a third axis (z-axis) perpendicular to x-y plane and passing through the origin. The Cartesian coordinates are called the rectangular coordinates.

## Polar Coordinates:

Some times, it is convenient to specify a point in space by its distance from the origin (r-value) and a value (θ-value)denotes the angle between the x-axis and *r*. $\theta$ is usually measured counterclockwise.




The Cartesian coordinates are related to the polar coordinates as

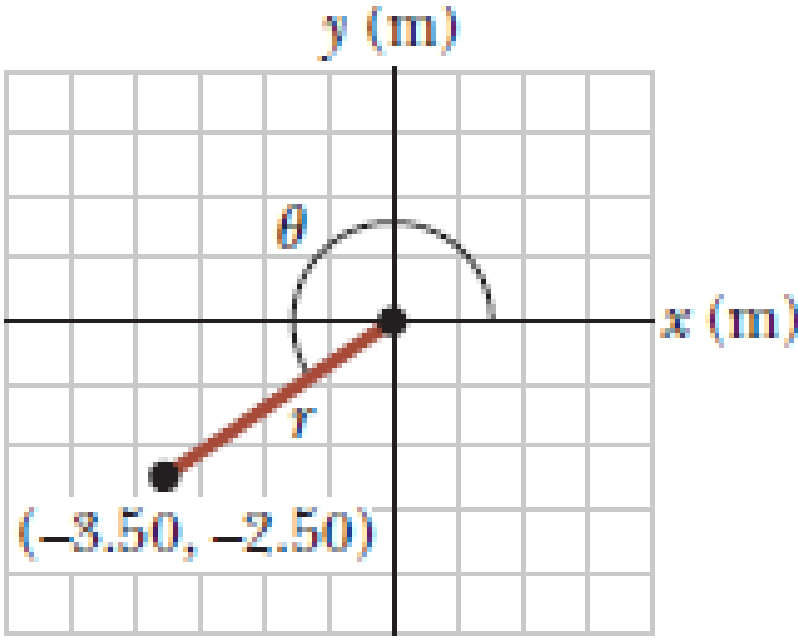
$$x = r\cos\theta, \qquad y = r\sin\theta$$

and $$r = \sqrt{x^2 + y^2},$$

with $$\tan\theta = \frac{y}{x}$$

These four expressions relating the coordinates (*x*, *y*) to the coordinates (*r*, $\theta$) apply only when $\theta$ is defined as shown.

**Example 3.1** The Cartesian coordinates of a point in the *x*-*y* plane are (*x*, *y*) = (-3.50, -2.50) m as shown. Find the polar coordinates of this point.




**Solution** To find *r* and θ we have

$$r = \sqrt{x^2 + y^2} = \sqrt{(-3.50)^2 + (-2.50)^2}$$

$$or \quad r = 4.30\,\text{m}$$

$$and \qquad \tan\theta = \frac{y}{x} = \frac{-2.50}{-3.50} = 0.714$$

$$or \quad \theta = 216^{\circ}$$

The polar coordinates of the point is (4.30m, 216$^{o}$)

# 3.2 Vector and Scalar Quantities

Physical quantities are either vectors or scalars:

| | | |
|---|---|---|
| Scalar | is the physical quantity that has magnitude only. | e.g., time, volume, mass, density, energy, distance, temperature. |
| Vector | is the physical quantity that has both magnitude and direction. | e.g., velocity, acceleration, displacement, force, area. |

In the book, the vector quantity will be distinguished from the scalar quantity by typing it in boldface with an arrow over the letter, like $\vec{\boldsymbol{A}}$ or simply **A** . The magnitude of the vector quantity will be denoted by $|\vec{\boldsymbol{A}}|$ or simply the italic letter *A*. The magnitude of a vector is always a positive number.

Geometrically, The vector quantity is represented by a straight line and an arrow at one end of the line.

Tail of the vector  Head of the vector

## 3.3 Some Properties of Vectors

**Equality of Two Vectors** Any two vectors are said to be equal if they have the same magnitude and point in the same direction, regardless of their location.

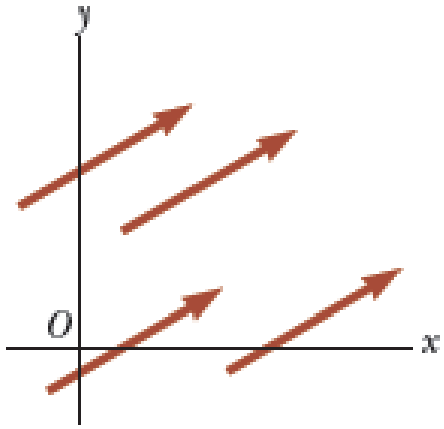
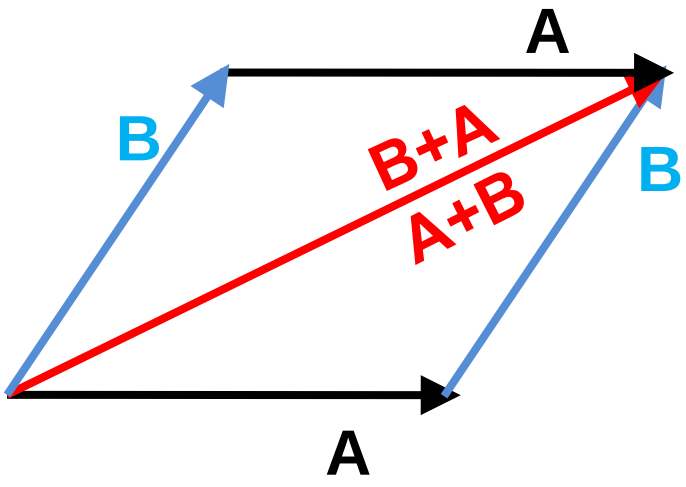



**Figure 3.5** These four vectors are equal because they have equal lengths and point in the same direction.

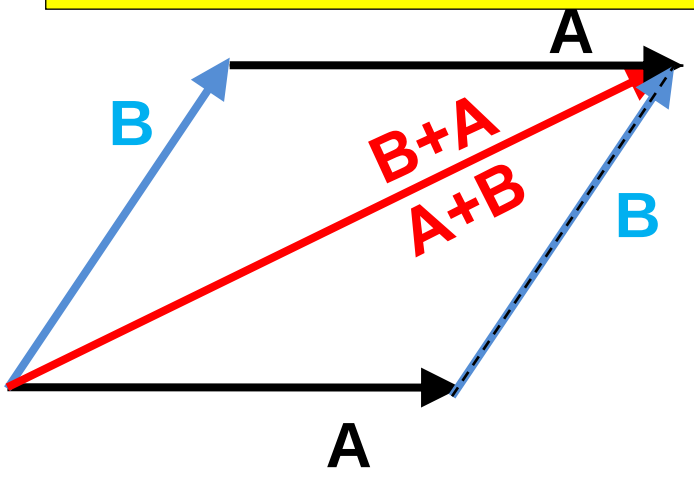


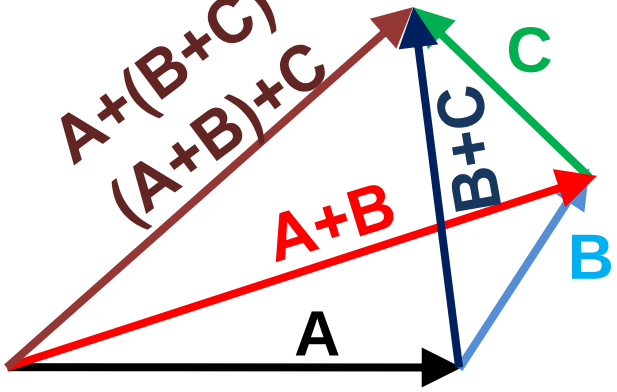
# Adding vectors (graphical method)

triangle method.




parallelogram method.



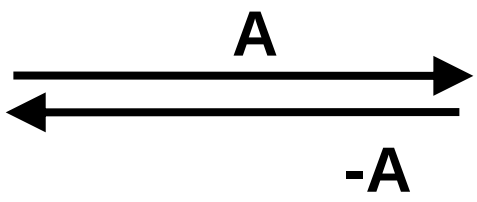




**A+B = B+A** (commutative)

**(A+B)+C = A+(B+C)** (associative)

# Negative of a Vectors

The vector $-\vec{A}$ is a vector with the same magnitude as the vector $\vec{A}$ but points in opposite direction.

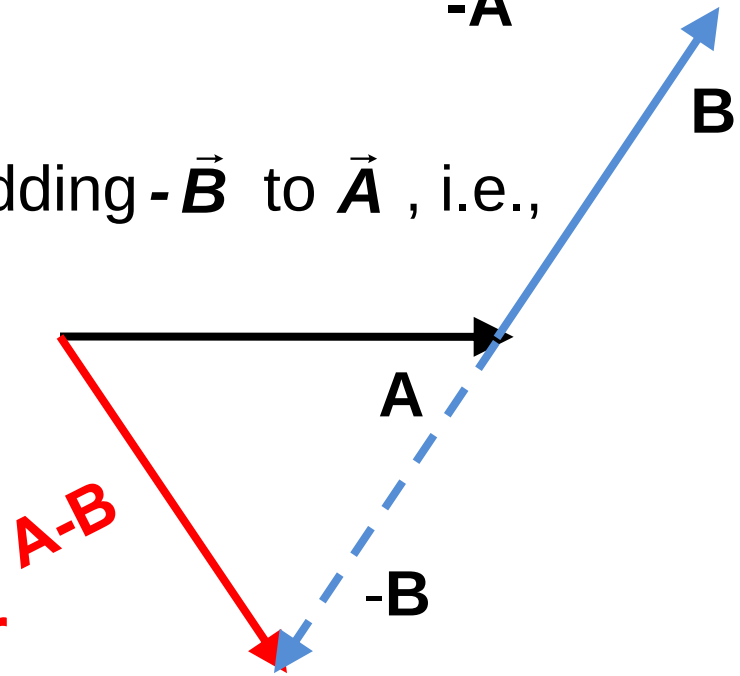



# Subtracting Vectors

Subtracting $\vec{B}$ from $\vec{A}$ is the same as adding $-\vec{B}$ to $\vec{A}$ , i.e.,

$$\vec{A} - \vec{B} = \vec{A} + \left(-\vec{B}\right)$$

Or $\vec{A} - \vec{B}$ is the vector when added to $\vec{B}$ must give $\vec{A}$




# Multiplying a Vector by a Scalar

If $\vec{A}$ is multiplied by a positive scalar $\lambda$, the product $\lambda\vec{A}$ Is a vector with the same direction as $\vec{A}$ and magnitude $\lambda A$. If $\lambda$ is negative the vector $\lambda\vec{A}$ points in the opposite direction of $\vec{A}$.

## • Quick Quiz (1)

The magnitudes of two vectors **A** and **B** are $A$ =12 units and $B$=8 units. Which pair of numbers represents the *largest* and *smallest* possible values for the magnitude of the resultant vector **R=A+B**?

a) 14.4 units, 4 units

b) 12 units, 8 units

c) 20 units, 4 units

d) Non of them

## • Quick Quiz (2)

Under what condition does the resultant vector $\boldsymbol{A} + \boldsymbol{B}$ have a magnitude equals to $A + B$?

a) ***A*** and ***B*** are parallel and in the same direction.

b) ***A*** and ***B*** are parallel and in opposite directions.

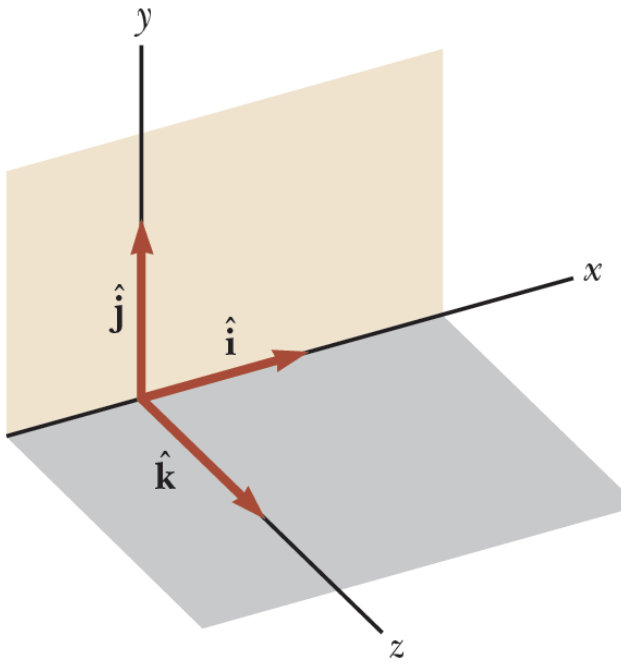
c) ***A*** and ***B*** are perpendicular.

d) Under no condition.

# Components of Vectors and Unit Vectors

A **unit vector** is a vector, with specific direction, having a magnitude of 1 without any units or dimensions.

In Cartesian coordinates three unit vectors are adopted to specify the positive directions of the three-axes.




$\hat{i}$ is directed along the positive *x*-axis

$\hat{j}$ is directed along the positive *y*-axis

$\hat{k}$ is directed along the positive *z*-axis

This means that if a vector **A** is directed a long the positive *x*-axis with a magnitude of *A*, this vector can be written as

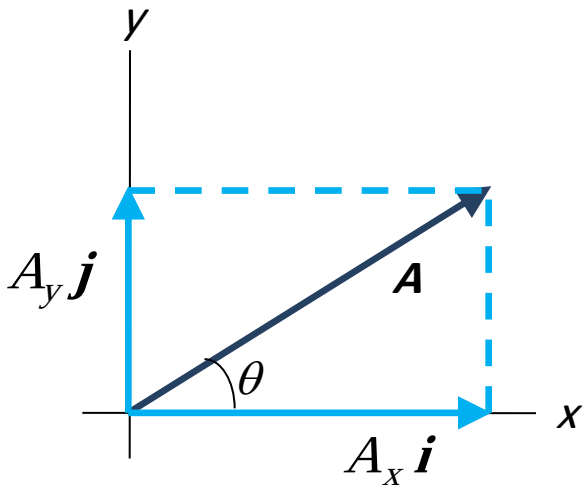
$$\vec{A} = A\hat{i}$$

The minus sign in front of any vector indicates the opposite direction of that vector, i.e. $-\hat{i}$ refers to the negative *x*-axis, and so for the other two unit vectors.

# Components of Vector

Consider the vector **A** in the *x*-*y* plane.

It is clear that the vector **A** can be represented by the sum of two vectors, one parallel to the *x*-axis ($A_x\,\hat{\boldsymbol{i}}$), and the other parallel to the *y*-axis ($A_y\,\hat{\boldsymbol{j}}$), i.e.,




$$\vec{\boldsymbol{A}} = A_x\hat{\boldsymbol{i}} + A_y\hat{\boldsymbol{j}}$$

$A_x$ and $A_y$ are called, respectively the *x*- and the *y*-components of the vector ***A***.

It is clear from the figure that

$$A_x = A\cos\theta \qquad \text{and} \qquad A_y = A\sin\theta$$

The magnitude of the vector is given by

$$A = |\mathbf{A}| = \sqrt{A_x^2 + A_y^2}$$

To find the direction of a vector it is enough to determine the angle $\theta$ the vector makes with a specific axis. The angle a vector makes with the x-axis is given by

$$\theta = \tan^{-1}\left(\frac{A_y}{A_x}\right)$$

In three dimensions any vector $\boldsymbol{A}$ can be resolved into three components as.

$$\vec{\boldsymbol{A}} = A_x\hat{\boldsymbol{i}} + A_y\hat{\boldsymbol{j}} + A_z\hat{\boldsymbol{k}}$$

Where $A_z$ is called the $z$-components of the vector $\boldsymbol{A}$.

Note that if the vector $\boldsymbol{A}$ is zero then all its components are zero.

# Adding Vectors

Now to add two vectors analytically proceed as follow:

(i) Resolve each vector into its components according to suitable coordinate axes.

(i) Add, algebraically, the *x*-components of the individual vectors to obtain the *x*-component of the resultant vector. Do the same thing for the other components, i.e., if

$$\vec{\boldsymbol{A}} = A_{\mathrm{x}}\hat{\boldsymbol{i}} + A_{\mathrm{y}}\hat{\boldsymbol{j}} + A_{\mathrm{z}}\hat{\boldsymbol{k}}$$

and $\vec{\boldsymbol{B}} = B_{\mathrm{x}}\hat{\boldsymbol{i}} + B_{\mathrm{y}}\hat{\boldsymbol{j}} + B_{\mathrm{z}}\hat{\boldsymbol{k}}$

Then the resultant vector $\boldsymbol{R}$ is

$$\vec{\boldsymbol{R}} = \vec{\boldsymbol{A}} + \vec{\boldsymbol{B}} = (A_{\mathrm{x}} + B_{\mathrm{x}})\hat{\boldsymbol{i}} + (A_{\mathrm{y}} + B_{\mathrm{y}})\hat{\boldsymbol{j}} + (A_{\mathrm{z}} + B_{\mathrm{z}})\hat{\boldsymbol{k}}$$

# • Quick Quiz (3)

For which of the following vectors is the magnitude of the vector equal to one of the components of the vector?

a) $\vec{A} = 2\hat{i} + 5\hat{j} + \hat{k}$

b) $\vec{A} = 2\hat{i} + 5\hat{j}$

c) $\vec{A} = -3\hat{j}$

d) $\vec{A} = 5\hat{k}$

**Example 3.3** Find the sum of the 2-displacement vectors

$$\vec{A} = (2.0\hat{i} + 2.0\hat{j})\text{m} \qquad \vec{B} = (2.0\hat{i} - 4.0\hat{j})\text{m}$$

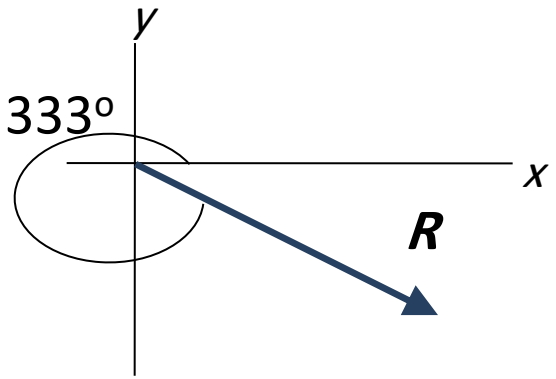
**Solution** From the given information we have

$$\vec{R} = \vec{A} + \vec{B} = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} \Rightarrow$$

$$\vec{R} = (2.0 + 2.0)\hat{i} + (2.0 - 4.0)\hat{j} = (4\hat{i} - 2.0\hat{j})\text{m}$$

$$|\vec{R}| = \sqrt{R_x^2 + R_y^2} = \sqrt{(4.0)^2 + (-2.0)^2} = 4.5\text{m}$$

$$\theta = \tan^{-1}\left(\frac{R_y}{R_x}\right) = \tan^{-1}\left(\frac{-2.0}{4.0}\right) = -27^o$$



**Example 3.4** A particle undergoes three consecutive displacements

$\vec{A} = (15\hat{i} + 30\hat{j} + 12\hat{k})$ cm $\vec{B} = (23\hat{i} - 14\hat{j} - 5.0\hat{k})$ cm $\vec{C} = (-13\hat{i} + 15\hat{j})$ cm

Find the resultant displacement and its magnitude.

**Solution**

$$\vec{R} = \vec{A} + \vec{B} + \vec{C} = (A_x + B_x + C_x)\hat{i} + (A_y + B_y + C_y)\hat{j} + (A_z + B_z + C_z)\hat{k} \Rightarrow$$

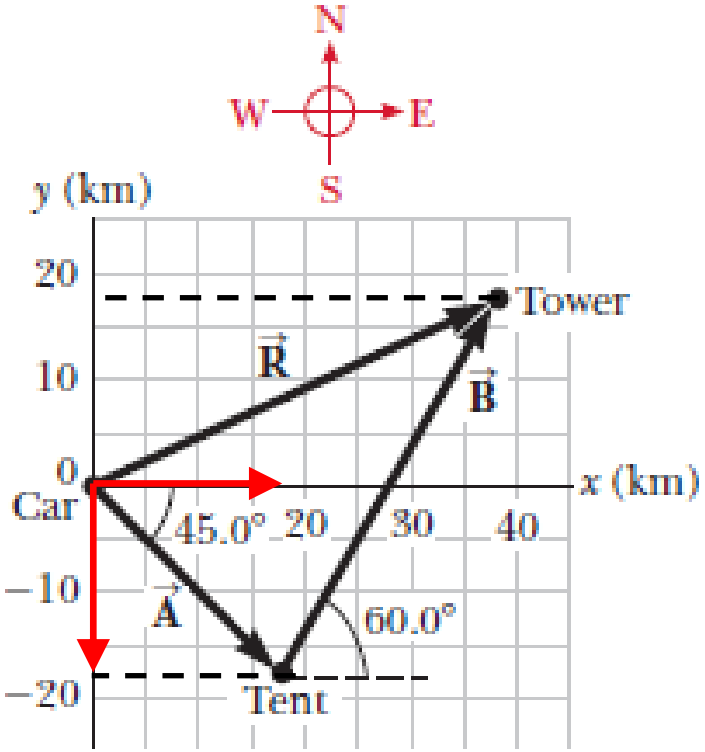
$$\vec{R} = (15 + 23 - 13)\hat{i} + (30 - 14 + 15)\hat{j} + (12 - 5.0 + 0)\hat{j} \Rightarrow$$

$$\vec{R} = (25\hat{i} + 31\hat{j} + 7.0\hat{j})\ \text{cm}$$

$$|\vec{R}| = \sqrt{R_x^2 + R_y^2 + R_z^2}$$

$$|\vec{R}| = \sqrt{(25.0)^2 + (31)^2 + (7.0)^2} = 40\,\text{cm}$$

**Example 3.5** A hiker begins a trip by first walking 25.0 km southeast from her car. She stops and sets up her tent for the night. On the second day, she walks 40.0 km in a direction 60.0° north of east, at which point she discovers a forest ranger's tower.
a) Determine the components of each displacement.
b) Find the resultant displacement in terms of unit vectors.




**Solution a)** The components of the first displacement are

$$A_x = A\cos\theta = 25.0\cos\left(-45^o\right) = 17.7\,\text{km}$$

$$A_y = A\sin\theta = 25.0\sin\left(-45^o\right) = -17.7\,\text{km}$$

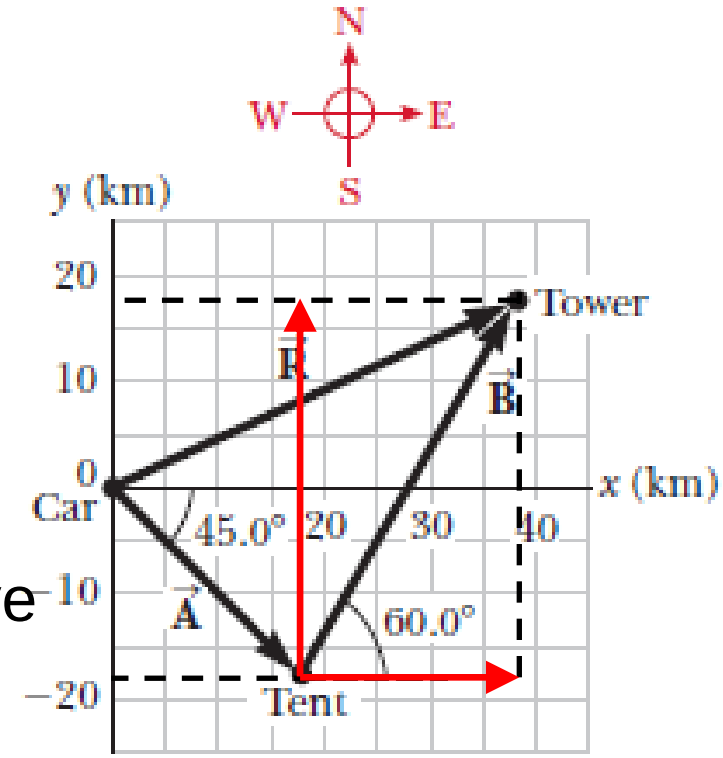
$$\Rightarrow \vec{A} = \left(17.7\hat{i} - 17.7\hat{j}\right)\text{km}$$

And the components of the second displacement are

$$B_x = B\cos\theta = 40.0\cos\left(60^o\right) = 20.0\,\text{km}$$

$$B_y = B\sin\theta = 40.0\sin\left(60^o\right) = 34.6\,\text{km}$$

$$\Rightarrow \vec{\boldsymbol{B}} = \left(20.0\hat{\boldsymbol{i}} + 34.6\,\hat{\boldsymbol{j}}\right)\text{km}$$




b) For the resultant displacement we have

$$\vec{\boldsymbol{R}} = \vec{\boldsymbol{A}} + \vec{\boldsymbol{B}} = (A_x + B_x)\hat{\boldsymbol{i}} + (A_y + B_y)\hat{\boldsymbol{j}} \Rightarrow$$

$$\vec{\boldsymbol{R}} = (20.0 + 17.7)\hat{\boldsymbol{i}} + (34.6 - 17.7)\hat{\boldsymbol{j}} \Rightarrow$$

$$\vec{\boldsymbol{R}} = \left(37.7\,\hat{\boldsymbol{i}} + 17.0\,\hat{\boldsymbol{j}}\right)\text{km}$$

To find the direction of the resultant displacement we have

$$\theta = \tan^{-1}\left(\frac{R_y}{R_x}\right) = \tan^{-1}\left(\frac{17.7}{37.7}\right) = 24.3^o$$

# Product of Vectors

## 1- Multiplying a vector by a scalar:

If $\lambda$ is a scalar and ***A*** is a vector then

$$\lambda\vec{\boldsymbol{A}} = \lambda A_x \hat{\boldsymbol{i}} + \lambda A_y \hat{\boldsymbol{j}} + \lambda A_z \hat{\boldsymbol{k}}$$

Note that the direction of $\lambda$ ***A*** is the same as the direction of ***A*** if $\lambda$ is +ve a and opposite to A if $\lambda$ is –ve.

## 2- Scalar (dot) product:

The scalar product of two vectors ***A*** and ***B***, denoted by $\vec{\boldsymbol{A}}\cdot\vec{\boldsymbol{B}}$ , is a **scalar** quantity defined by,

$$\vec{\boldsymbol{A}}\cdot\vec{\boldsymbol{B}} = \left|\vec{\boldsymbol{A}}\right|\left|\vec{\boldsymbol{B}}\right|\cos\theta$$

with $\theta$ is the smallest angle between the two vectors.

Since ***i***, ***j***, and ***k*** are unit vectors perpendicular to each other, and using the last equation we can write

$$\hat{i}\cdot\hat{j}=\hat{i}\cdot\hat{k}=\hat{k}\cdot\hat{j}=0 \qquad \hat{i}\cdot\hat{i}=\hat{j}\cdot\hat{j}=\hat{k}\cdot\hat{k}=1$$

$$\vec{A}\cdot\vec{B}=\left(A_x\hat{i}+A_y\hat{j}+A_z\hat{k}\right)\cdot\left(B_x\hat{i}+B_y\hat{j}+B_z\hat{k}\right)$$

$$=A_xB_x\hat{i}\cdot\hat{i}+A_xB_y\hat{i}\cdot\hat{j}+A_xB_z\hat{i}\cdot\hat{k}$$

$$+A_yB_x\hat{j}\cdot\hat{i}+A_yB_y\hat{j}\cdot\hat{j}+A_yB_z\hat{j}\cdot\hat{k}$$

$$+A_zB_x\hat{k}\cdot\hat{i}+A_zB_y\hat{k}\cdot\hat{j}+A_zB_z\hat{k}\cdot\hat{k}$$

$$\Rightarrow \quad \vec{A}\cdot\vec{B}=A_xB_x+A_yB_y+A_zB_z$$

**Example 7.2** You are given the vectors $\vec{A}=2\hat{i}+3\hat{j}$ $\vec{B}=-\hat{i}+2\hat{j}$

a) Find the dot product between the two vectors.
b) Find the angle between the two vectors.

**Solution a)** $\vec{A}\cdot\vec{B}=A_xB_x+A_yB_y+A_zB_z=2(-1)+3(2)=-2+6=4$

b) To find the angle between the two vectors we have

$$\vec{A} = 2\hat{i} + 3\hat{j} \qquad \vec{B} = -\hat{i} + 2\hat{j}$$

$$\vec{A}\cdot\vec{B} = |\vec{A}||\vec{B}|\cos\theta \quad \Rightarrow \cos\theta = \frac{\vec{A}\cdot\vec{B}}{|\vec{A}||\vec{B}|}$$

$$\text{But} = |\vec{A}| = \sqrt{2^2 + 3^2} = \sqrt{13} \qquad \text{and} \qquad |\vec{B}| = \sqrt{(-1)^2 + 2^2} = \sqrt{5}$$

$$\Rightarrow \cos\theta = \frac{4}{\sqrt{13\times 5}} = 0.5 \qquad \Rightarrow \theta = 60^{o}$$

- **Quick Quiz (4)**

Which of the following statements is true?

a) $\vec{A}\cdot\vec{B}$ is larger than *AB*

b) $\vec{A}\cdot\vec{B}$ is smaller than *AB*

c) $\vec{A}\cdot\vec{B}$ could be larger than or smaller than *AB*

d) $\vec{A}\cdot\vec{B}$ could be equal to *AB*

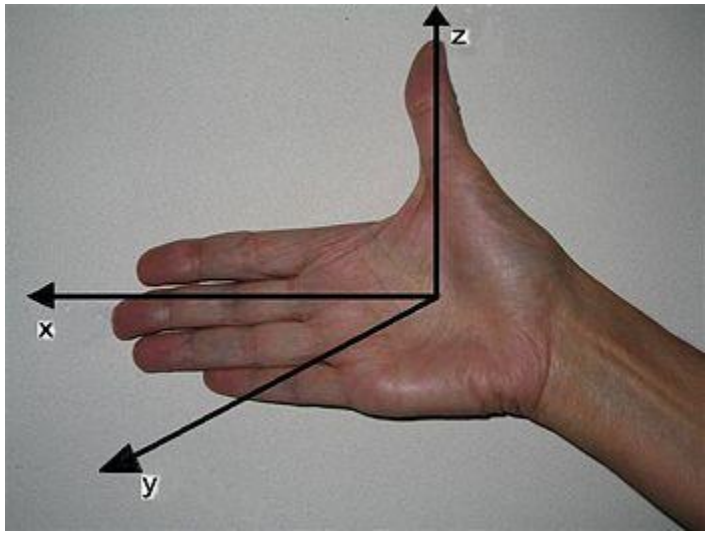
## 3- Vector (cross) product:

The vector product of two vectors ***A*** and ***B***, written as ***A***×***B***, is a third vector ***C*** with a magnitude given by

$$\left|\vec{C}\right| = \left|\vec{A}\times\vec{B}\right| = \left|\vec{A}\right|\left|\vec{B}\right|\sin\theta$$

The vector ***C*** is perpendicular to the plane of ***A*** and ***B***. Since there are two directions perpendicular to a given plane, the **right hand rule** is used to decide to which direction the vector ***C*** = ***A***×***B*** is directed.

The rule states that the four fingers of the right hand are pointed along ***A*** and then curled toward ***B*** through the smaller angle between ***A*** and ***B***. The thumb then gives the direction of ***C***.




Using this rule it is clear that.

$$\vec{A}\times\vec{B} = -\vec{B}\times\vec{A}$$

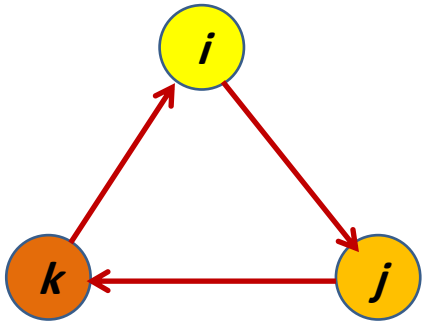
By the same rule one can verify that

$$\hat{i}\times\hat{i} = \hat{j}\times\hat{j} = \hat{k}\times\hat{k} = 0$$

$$\hat{i}\times\hat{j} = \hat{k}$$

$$\hat{j}\times\hat{k} = \hat{i}$$

$$\hat{k}\times\hat{i} = \hat{j}$$




$$\vec{A}\times\vec{B} = \left(A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}\right)\times\left(B_x\mathbf{i} + B_y\mathbf{j} + B_z\mathbf{k}\right)$$

$$= A_xB_x\mathbf{i}\times\mathbf{i} + A_xB_y\mathbf{i}\times\mathbf{j} + A_xB_z\mathbf{i}\times\mathbf{k}$$

$$+ A_yB_x\mathbf{j}\times\mathbf{i} + A_yB_y\mathbf{j}\times\mathbf{j} + A_yB_z\mathbf{j}\times\mathbf{k}$$

$$+ A_zB_x\mathbf{k}\times\mathbf{i} + A_zB_y\mathbf{k}\times\mathbf{j} + A_zB_z\mathbf{k}\times\mathbf{k}$$

$$= (A_yB_z - A_zB_y)\mathbf{i} + (A_zB_x - A_xB_z)\mathbf{j} + (A_xB_y - A_yB_x)\mathbf{k}$$

Or equivalently we have

$$\vec{A}\times\vec{B}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ A_x & A_y & A_z\\ B_x & B_y & B_z\end{vmatrix}=(A_yB_z-A_zB_y)\hat{i}+(A_zB_x-A_xB_z)\hat{j}+(A_xB_y-A_yB_x)\hat{k}$$

**Example 11.1** You are given the vectors $\vec{A}=2\hat{i}+3\hat{j}$ $\vec{B}=-\hat{i}+2\hat{j}$

a) Find the cross product between the two vectors.
b) Verify that $\vec{A}\times\vec{B}=-\vec{B}\times\vec{A}$

**Solution a)** $$\vec{A}\times\vec{B}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ A_x & A_y & A_z\\ B_x & B_y & B_z\end{vmatrix}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 3 & 0\\ -1 & 2 & 0\end{vmatrix}$$

$$=(3\times0-0\times2)\hat{i}+(0\times-1-2\times0)\hat{j}+(2\times2-3\times-1)\hat{k}=7\hat{k}$$

b) $$\vec{B}\times\vec{A}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ B_x & B_y & B_z\\ A_x & A_y & A_z\end{vmatrix}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ -1 & 2 & 0\\ 2 & 3 & 0\end{vmatrix}$$

$$=(2\times0-0\times3)\hat{i}+(0\times2--1\times0)\hat{j}+(-1\times3-2\times2)\hat{k}\quad=-7\hat{k}$$

- **Quick Quiz (5)**

Witch of the following is always zero?

a) $(\boldsymbol{A}\times\boldsymbol{B})\times\boldsymbol{A}$

b) $(\boldsymbol{A}\times\boldsymbol{B})\cdot\boldsymbol{A}$

c) $(\boldsymbol{A}\times\boldsymbol{B})\times\boldsymbol{B}$

d) $(\boldsymbol{A}\cdot\boldsymbol{B})\times\boldsymbol{A}$